TINMORRY

Technology Data Sheet (TDS)

TINMORRY Matte PLA+

Version 1.0

• Basic Introduction

TINMORRY Matte PLA+ is an upgraded version of matte PLA with optimized formula and better overall performance. It has the ease of printing (wide machine compatibility range and high molding quality) and biodegradability of PLA, as well as the matte surface and uniform gloss of generic matte PLA. In addition, compared with generic matte PLA, it also has higher printing speed (up to 330 mm/s at 220 ℃ nozzle temp) and higher layer strength, and is less likely to occur in defects such as oozing and stringing. Therefore, TINMORRY Matte PLA+ is a kind of basic and commonly-used filament that is highly competitive.

• Specifications

Subjects	Data
Diameter	1.75 ± 0.03 mm
Net filament weight	1000 g
Length	300 ± 5 m
Spool material	ABS ((Temperature resistance: 80 °C)
Spool size	External Diameter: 198 mm; Inner Diameter: 56 mm; Width: 65 mm

• Recommended Printing Settings

Subjects	Data
Drying settings before printing	50 - 60 °C，6 - 8 h (Blast drying oven, or filament drying box)
Printing temperature and humidity	≤ 30 ℃, ≤ 25% RH （Put in an air-tight container and sealed with desiccant）

Storage temperature and humidity	≤ 30 °C, ≤ 20% RH（Sealed with desiccant）
Compatible support material	Itself or support filament for PLA
Compatible printer type	Open-frame, enclosed-frame
Compatible nozzle material	Brass, stainless steel, hardened steel, etc
Compatible nozzle size	0.2, 0.4, 0.6, 0.8 mm
Compatible plate type	Textured PEI Plate or other plate
Plate surface preparation	Glue
Nozzle temperature	200 - 230 °C
Bed temperature	60 - 75 °C (depending on the printer and plate types)
Max Printing speed	300 - 330 mm/s（220 °C，Bambu X1C, P1S, A1）
Chamber temperature	< 40 °C

• Properties

The commonly used physical properties, mechanical properties, chemical properties, printing properties and other properties of TINMORRY Matte PLA+ have been tested and are shown in the tables below.

• Physical Properties

Subjects	Testing Methods	Data
Density	ISO 1183	1.39 g/cm^3
Saturated water absorption rate	25 °C , 55% RH	0.43%
Melt index	210 °C, 2.16 kg	7.6 ± 1.2 g/ 10 min
Glass transition temperature	DSC, 10 °C/min	54 °C
Crystallization temperature	DSC, 10 °C/min	N / A
Melting temperature	DSC, 10 °C/min	153 °C
Vicar softening temperature	ISO 306, GB/T 1633	59 °C
Heat deflection temperature 1	ISO 75, 1.82 MPa	50 °C
Heat deflection temperature 2	ISO 75, 0.45 MPa	56 °C

• Mechanical Properties

Properties	Testing Methods	Data
Tensile strength (XY)	ISO 527, GB/T 1040	34 ± 3 MPa
Tensile strength (Z)	ISO 527, GB/T 1040	21 ± 4 MPa
Young's modulus (XY)	ISO 527, GB/T 1040	2637 ± 183 MPa
Young's modulus (Z)	ISO 527, GB/T 1040	2085 ± 217 MPa
Breaking elongation rate (XY)	ISO 527, GB/T 1040	6.1% ± 1.6%
Breaking elongation rate (Z)	ISO 527, GB/T 1040	2.2% ± 0.7%
Bending strength (XY)	ISO 178, GB/T 9341	66 ± 3 MPa
Bending strength (Z)	ISO 178, GB/T 9341	37 ± 2 MPa
Bending modulus (XY)	ISO 178, GB/T 9341	2936 ± 247 MPa
Bending modulus (Z)	ISO 178, GB/T 9341	2560 ± 162 MPa

Impact strength (XY)	ISO 179, GB/T 1043	34.5 ± 2.7 kJ/m²
Impact strength (Z)	ISO 179, GB/T 1043	8.2 ± 1.6 kJ/m²

• Chemical and Other Properties

Subjects	Data
Component	PLA
Skin irritation	No
Resistance to water	Yes
Resistance to oil and grease	Resistant to most kinds of daily oil and grease
Resistance to organic solvent	Not resistant to some organic solvents
Resistance to weak acid	Yes for a short time
Resistance to strong acid	Not resistant
Resistance to weak alkali	Not resistant
Resistance to strong alkali	Not resistant
Flammability	Yes
Odor during printing	Odorless
Odor during combustion	Slightly pungent

• Specimen Info

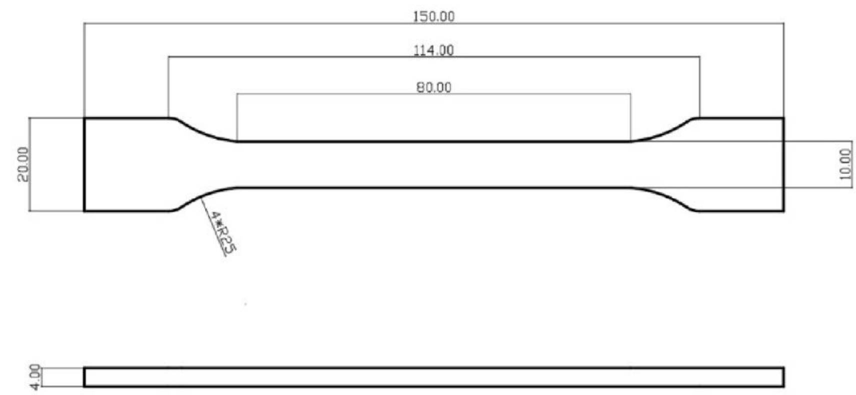
Subjects	Data
Size	Shown at the following pictures
Nozzle temperature	220 °C
Bed temperature	55 °C
Printing speed	XY: 270 mm/s; Z: 157 mm/s
Infill density	100%
Infill pattern	Concentric

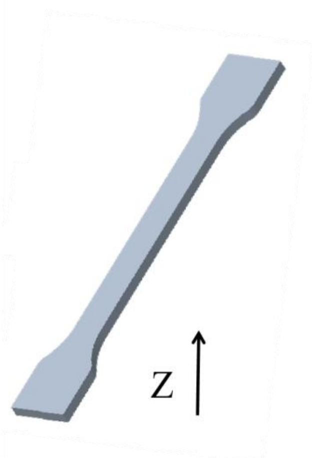
Statements：

1. The test specimens were printed under the key parameters mentioned above. Before testing, all specimens were placed in an indoor environment at about 25 °C for 12 hours. Please note that different ambient temperatures, fan speeds and layer times can lead to varying cooling, which can affect the mechanical properties of the specimens, especially those in the Z-direction. Additionally, different printers can also result in variations in printing quality.
2. If high printing quality is required, it is recommended to dry every kind of TINMORRY filaments before printing and use an air-tight container and desiccant to protect them from moisture during the printing process (with humidity inside the container less than 25%) to reduce the risk of issues related to moisture absorption in filaments. The recommended drying conditions are shown above. Note that when drying them, avoid placing the spools near the heater to prevent damage to the spools or the filaments due to excessive temperature. Do not use a kitchen oven or microwave oven.
3. After annealing, the strength, stiffness and heat resistance of TINMORRY Matte PLA+ prints can be slightly increased. However, the toughness can be increased, and some prints may undergo shrinkage or deformation after the annealing process. Note that the annealing effect is also affected by the size, structure, wall thickness, and infill density of the prints

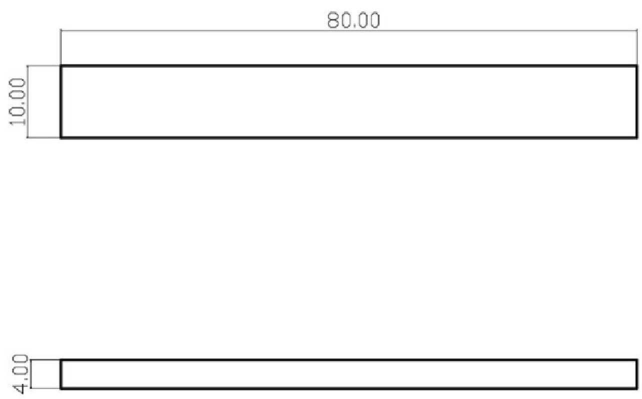
themselves. If you want to anneal them, it is recommended to use a blast drying oven which has uniform temperature, and it is recommended to first anneal at about 50 °C for about 4 hours, then slowly increase the temperature to about 55 °C and keep it for about 4 to 6 hours; if the prints are significantly deformed halfway, it should be stopped immediately and a lower temperature should be used instead.

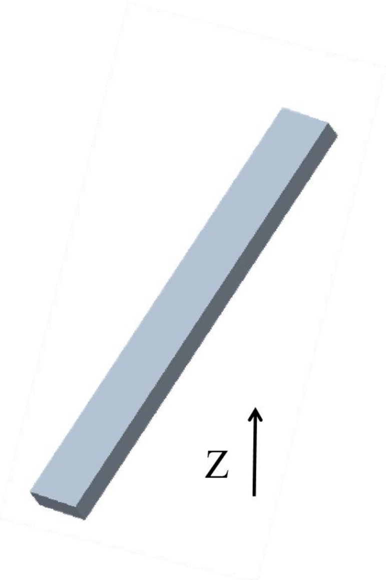
1. Tensile Test





2. Bending Test and Impact Test





• Disclaimer

The above properties data is obtained by TINMORRY through testing with standard samples, standard methods, and specific equipment, and is for reference and comparison only; if some data are changed, TINMORRY will update this TDS, but may not be able to inform the users, and please forgive us. The actual performance of 3D prints depends not only on the performance of the filaments used, but also on many factors, such as the degree of moisture absorption of the filaments, the printer, environmental conditions, model characteristics, printing parameters, etc.. When using TINMORRY FDM 3D printing filaments, users are responsible for the legality, safety, and performance of the prints. TINMORRY is responsible for the quality of our filaments, but not for any damage or injury that occurs during using them,or the usage scenarios of them.